Mohak Kapoor

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Technical Skills

Programming Languages	Python (Advanced), C++
DevOps / Backend	GitHub Actions, Docker, CI/CD, PostgreSQL, REST APIs
Machine Learning	Regression, Classification, Deep Learning, Time-Series Forecasting
Libraries / Frameworks	TensorFlow, scikit-learn, Pandas, NumPy, Django, Polars
Currently Exploring	Kubernetes, LangChain, LoRA, RAG, Kafka, Prompts

Projects

Cross-Market Index Forecasting Model (CMFM v1.0) *Jan–May 2025*

- Developed a deep learning model to predict index bid-open prices using 7M rows of 1-minute Dukascopy data, achieving MAPE of 3.1% (Japan), 4.2% (UK), and 12% (US) on 2025 test data.
- Built a scalable Python/Polars/TensorFlow pipeline, processing data in chunks, engineering features like midprice, MACD, lagged log returns, & z-score normalization and market session-aware null imputation.
- Implemented a CNN-LSTM model with self-attention, using masks for market closures and L1/L2 regularization, optimized for i9/32GB RAM with GPU memory management.

Nifty50 Trend Prediction

- Built a Random Forest model to predict Nifty50 up/down trends using Yahoo Finance data, achieving 68% accuracy (vs. 57% market baseline).
- Tools: Python, sklearn, Pandas, NumPy, yfinance

Solar Power Generation Predictor

- Developed a TensorFlow model to predict solar power output using OpenWeatherMap API, geographical, and time-based features, hosted on a Django web platform.
- Tools: Python, sklearn, TensorFlow, Pandas, NumPy, Django, OpenWeatherMap API

Experience

JPMorgan Chase & Co. Quantitative Research Virtual Experience *Jan 2025*

- Analyzed a book of loans to estimate customer probability of default, applying quantitative research methods in a simulated environment.
- Used dynamic programming to convert FICO scores into categorical data, enhancing default prediction model robustness.

HumanizeIQ — AI Intern Integrations *June 2025 – Present*

- Developed Model Context Protocol (MCP) servers using Cloudflare Workers for AI-powered image and document generation, with async processing and R2 bucket storage.
- Built scalable MCP infrastructure that returns immediate job IDs and processes generation requests in the background, implementing polling-based result retrieval.
- Integrated MCP tools with LLM chat systems, enabling seamless generation workflows for various document types with embedded AI-generated content.
- Implemented robust async job management with status tracking & result fetching for production AI tools.

Education

Jaypee Institute of Information Technology, Noida-62 *Sept 2022 – July 2026*

B.Tech in Electronics and Communication Engineering

- **Relevant Coursework:** Introduction to Computer Systems, Fundamentals of Computer Science, Multivariate Calculus, Differential & Integral Calculus, ML and Deep Learning, Control Systems, A in Advanced Statistics